

Taking Serene to the Cloud

Prathmesh, when you are ready to launch your application to real users, your local `http://localhost:11434` address will no longer work. You need a cloud server.

Because Large Language Models (LLMs) like Phi-4 or Llama 3 are mathematically heavy, a standard cheap web server (like a \$5 DigitalOcean droplet) will be too slow. You need a server with a **GPU** (Graphics Processing Unit).

Here are the two best ways to deploy your AI pipeline for production.

Option 1: Self-Host Ollama on a GPU Cloud (Full Control & Maximum Privacy)

If you want to keep using Ollama exactly as you are now, you will rent a virtual machine in the cloud that has a dedicated GPU, install Ollama on it, and point your backend code to that new IP address.

- **Best Providers for this:**
 - **RunPod or Lambda Labs:** These are built specifically for AI. You can rent a machine with an Nvidia RTX 3090 or A4000 for about \$0.30 to \$0.50 per hour.
 - **AWS (Amazon Web Services):** You would use an "EC2 g4dn" instance. It's more enterprise-grade but slightly more complex to set up.
- **How it works:**
 1. You rent the server and log into it.
 2. You run the exact same command you ran on your laptop: `curl -fsSL https://ollama.com/install.sh | sh`
 3. You run `ollama run phi4`.
 4. You expose the server's port `11434` to the internet (using a secure tunnel or Nginx reverse proxy).
 5. In your `insights_agent.py` script, you change `http://localhost:11434/v1` to `http://<YOUR_CLOUD_SERVER_IP>:11434/v1`.

Option 2: Serverless Open-Source APIs (The "Easy" Production Route)

Managing your own GPU server can get expensive if users aren't using the app 24/7, because you pay by the hour even when the server is sitting idle.

Instead of hosting Ollama yourself, you can use a **Managed API Provider** that hosts open-source models (like Llama 3 or Mixtral) for you. You only pay for the exact tokens (words) generated, which is usually fractions of a cent per request.

- **Best Providers for this:**
 - **Groq:** Insanely fast (generates hundreds of words per second). Great for chatbots.
 - **Together AI or Fireworks AI:** Excellent for hosting open-source models securely.
- **How it works:**
 1. You create an account on Together AI or Groq and get a real API key.

2. Because your `insights_agent.py` code already uses the `instructor` library and the `OpenAI` client format, the code change is literally just two lines.

Example Code Change for Option 2:

```
# Old Local Ollama Code:
client = instructor.from_openai(
    openai.OpenAI(base_url="http://localhost:11434/v1", api_key="ollama"),
    mode=instructor.Mode.JSON,
)

# New Cloud Production Code (e.g., using Together AI):
client = instructor.from_openai(
    openai.OpenAI(
        base_url="https://api.together.xyz/v1",
        api_key="your_real_production_api_key"
    ),
    mode=instructor.Mode.JSON,
)
```

The Recommended Path for Serene

Start with **Option 2 (Serverless APIs)** for your MVP. Services like Groq or Together AI are incredibly cheap, require zero DevOps or server maintenance, and use the exact same open-source models you are testing locally. Once Serene hits thousands of daily active users, you can transition to **Option 1 (Self-hosting on AWS/RunPod)** to lock down your data pipeline completely.

